Theorem 3.2.1 **(Existence and Uniqueness Theorem)**

Consider the initial value problem → 2nd order, linear ODE

$$\mathbf{1}\ y'' + p(t)y' + q(t)y = g(t), \qquad y(t_0) = y_0, \quad y'(t_0) = y_0' \tag{4}$$

where p, q, and g are continuous on an open interval I that contains the point t_0. Then there is exactly one solution $y = \phi(t)$ of this problem, and the solution exists throughout the interval I.

We emphasize that the theorem says three things:

1. The initial value problem *has* a solution; in other words, a solution exists.
2. The initial value problem has *only one* solution; that is, the solution is *unique*.
3. The solution ϕ is defined *throughout the interval* I where the coefficients are continuous and is at least twice differentiable there.

In each of Problems 7 through 12, determine the longest interval in which the given initial value problem is certain to have a unique twice-differentiable solution. Do not attempt to find the solution.

3.2. 8. $(t-1)y'' - 3ty' + 4y = \sin t, \quad y(-2) = 2, \quad y'(-2) = 1$

12. $(x-2)y'' + y' + (x-2)(\tan x)y = 0, \quad y(3) = 1, \quad y'(3) = 2$

8) $y'' - \underbrace{\frac{3t}{t-1}}_{p(t)} y' + \underbrace{\frac{4}{t-1}}_{q(t)} y = \underbrace{\frac{\sin t}{t-1}}_{g(t)}$ $\qquad t_0 = -2$

discts at $t=1$ $\qquad I = (-\infty, 1)$ $(1, \infty)$

by Ex & Un. Thm.

12) $y'' + \underbrace{\frac{1}{x-2}}_{p(x)} y' + \underbrace{(\tan x)}_{q(x)} y = 0$ $\qquad t_0 = 3 \Rightarrow I = \left(2, \frac{3\pi}{2}\right)$

discts at $x=2$ $\qquad$ discts at $x = \frac{\pi}{2} + k\pi \quad k \in \mathbb{Z}$.

$$\cdots \cup \left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right) \cup \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, 2\right) \cup \left(2, \frac{3\pi}{2}\right) \cup \left(\frac{3\pi}{2}, \frac{5\pi}{2}\right) \cdots$$

We talked about Wronskian and reviewed thms 3.2.2, 3.2.3, 3.2.4 and their proofs. (Read p. 148-150 on 10th Ed.)

18. If the Wronskian W of f and g is t^2e^t, and if $f(t) = t$, find $g(t)$.

$$W(f,g)(t) = \begin{vmatrix} f & g \\ f' & g' \end{vmatrix} = (fg' - gf')(t) = t^2e^t$$

$f(t)=t \Rightarrow \quad tg' - g = t^2e^t$

$g_h(t) = c \cdot \frac{1}{t}$ — 1st order, linear ODE

$g(t) = te^t + ct$ — $\Rightarrow$ integrating factor $\mu(t)$

In each of Problems 24 through 27, verify that the functions y_1 and y_2 are solutions of the given differential equation. Do they constitute a fundamental set of solutions?

3.2. 24. $y'' + 4y = 0$; $y_1(t) = \cos 2t$, $y_2(t) = \sin 2t$ $\quad y(0)=1 \quad y'(0)=-1$ $\quad \{y_1(t), y_2(t)\}$

$$W(y_1,y_2)(t) = \begin{vmatrix} \cos 2t & \sin 2t \\ -2\sin 2t & 2\cos 2t \end{vmatrix} = 2(\cos^2 2t + \sin^2 2t) = 2 \quad e^t$$

YES.

$W(t) \neq 0$

discussion:

$W(y_1,y_2)(t) = \begin{vmatrix} \cdots & \cdots \\ \cdots & \cdots \end{vmatrix} = \cdots = t \quad \rightarrow$ for $t_0 = 1$ $W = 1$ YES.

(here, $p(t) \sim \frac{1}{t}$) — only one point where $W \neq 0$ is enough

$W(t) \equiv 0$ $\quad = t - t = 0 \quad \rightarrow$ NO.

$y'' + 4y = 0 \qquad y_1 = \cos 2t \qquad y_2 = \sin 2t$

$y = c_1 \cos 2t + c_2 \sin 2t$

$y' = c_1(-2\sin 2t) + c_2(2\cos 2t)$

After Abel's Thm we see, if such a Wronskian exists, then all points to where $W(t_0)=0$ are excluded somehow. (for example, $p(x)$ is undefined for those t_0's etc.)

$y(0) = 1 \Rightarrow \quad c_1\cos(0) + c_2\sin(0) = 1$

$y'(0) = -1 \Rightarrow \quad c_1(-2\sin(0)) + c_2(2\cos(0)) = -1$

$$\begin{bmatrix} \cos(0) & \sin(0) \\ -2\sin(0) & 2\cos(0) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Theorem 3.2.7 (Abel's Theorem)[7]

If y_1 and y_2 are solutions of the differential equation

$$L[y] = y'' + p(t)y' + q(t)y = 0, \tag{22}$$

where p and q are continuous on an open interval I, then the Wronskian $W(y_1, y_2)(t)$ is given by

$$W(y_1, y_2)(t) = c \exp\left[-\int p(t)\,dt\right], \tag{23}$$

where c is a certain constant that depends on y_1 and y_2, but not on t. Further, $W(y_1, y_2)(t)$ either is zero for all t in I (if $c = 0$) or else is never zero in I (if $c \neq 0$).

Recall the ExUn. Thm 3.2.1!

In each of Problems 29 through 32, find the Wronskian of two solutions of the given differential equation without solving the equation.

3.2.30. $(\cos t)y'' + (\sin t)y' - ty = 0$

Abel

$$W(y_1, y_2)(t) = c\,e^{-\int p(t)\,dt} = c_1 e^{-\int \frac{\sin t}{\cos t}\,dt} = c\,e^{\ln|\cos t| + c_1} = c(\cos t)$$

undefined if $\cos t = 0$

$= 0$ if $\cos t = 0$

3.2.34. If the differential equation $ty'' + 2y' + te^t y = 0$ has y_1 and y_2 as a fundamental set of solutions and if $W(y_1, y_2)(1) = 2$, find the value of $W(y_1, y_2)(5)$.

Abel

$$W(y_1, y_2)(t) = c\,e^{-\int p(t)\,dt} = c\,e^{-\int \frac{2}{t}\,dt} = c\,e^{-2\ln|t| + c} = c\,e^{\ln t^{-2}} = c\,t^{-2}$$

$W(1) = 2 \Rightarrow c(1)^{-2} = 2 \Rightarrow c = 2$

$W(5) = ? = 2 \cdot (5)^{-2} = 2/25.$

In each of Problems 7 through 16, find the general solution of the given differential equation.

3.3.7. $y'' - 2y' + 2y = 0$

2nd order, linear ODE & hom. w/ const. coefs.

char eqn: $r^2 - 2r + 2 = 0 \Rightarrow r_{1,2} = 1 \pm i \rightarrow y_1 = e^t \sin t$, $y_2 = e^t \cos t$

$(r-1)^2 + 1 = 0$

$y = e^{rt}$

$e^{(1+i)t} = e^t \cdot e^{it} = e^t(\cos t + i \sin t)$

$e^{(1-i)t} = e^t \cdot e^{-it} = e^t(\cos(-t) + i \sin(-t))$

$e^{i(\cdots)}$

$\cos t$ $\quad -\sin t$

Thm 3.2.6: $y = u(t) + iv(t) \Rightarrow u(t), v(t)$ soln.

$e^{ait} = \cos at + i \sin at = \text{cis}\, at$

$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \Rightarrow e^{ait} = \sum_{n=0}^{\infty} \frac{(ait)^n}{n!} = \sum_{n=0}^{\infty} (ai)^n \frac{t^n}{n!}$

$= \sum_{k=0}^{\infty} (i)^{2k} \frac{(at)^{2k}}{(2k)!} + \underbrace{\sum_{m=0}^{\infty} (i)^{2m+1} \frac{(at)^{2m+1}}{(2m+1)!}}_{i(\quad)}$

$= \underbrace{\sum_{k=0}^{\infty} (-1)^k \frac{(at)^{2k}}{(2k)!}}_{\cos at} + i \underbrace{\sum_{m=0}^{\infty} (-1)^m \frac{(at)^{2m+1}}{(2m+1)!}}_{\sin at}$

$= \cos at + i \sin at$

$$W(e^t \cos t, e^t \sin t) = \begin{vmatrix} e^t \cos t & e^t \sin t \\ e^t \cos t - e^t \sin t & e^t \sin t + e^t \cos t \end{vmatrix}$$

$$= e^{2t} \begin{vmatrix} \cos t & \sin t \\ \cos t - \sin t & \sin t + \cos t \end{vmatrix}$$

$= e^{2t} (\cos t \sin t + \cos^2 t - \cos t \sin t + \sin^2 t) = e^{2t}$

gen. soln:

$y = c_1 y_1 + c_2 y_2 = c_1 e^t \cos t + c_2 e^t \sin t$

$r = a \pm ib \rightarrow y_1 = e^{at} \cos bt \quad y_2 = e^{at} \sin bt$

$$W(y_1, y_2)(t) = \begin{vmatrix} e^{at} \cos bt & e^{at} \sin bt \\ ae^{at} \cos bt - e^{at} b \sin bt & ae^{at} \sin bt + e^{at} b \cos bt \end{vmatrix}$$

$$= e^{2at} \begin{vmatrix} \cos bt & \sin bt \\ a \cos bt - b \sin bt & a \sin bt + b \cos bt \end{vmatrix}$$

$$= e^{2at}\left(a\cos bt\sin bt + b\cos^2 bt - a\cos bt\sin bt + b\sin^2 bt\right)$$

$$= e^{2at}\, b \neq 0$$ → so $\{e^{at}\cos bt, e^{at}\sin bt\}$ always forms a fundamental set of solutions unless $b=0$ or $a=0$

In each of Problems 17 through 22, find the solution of the given initial value problem. Sketch the graph of the solution and describe its behavior for increasing t.

3.3. 17. $y'' + 4y = 0, \quad y(0) = 0, \quad y'(0) = 1$

$r^2 + 4 = 0$

2nd order, lin. ODE, hom & const. coef. ⇒ char eqn.

$r^2 = -4 \Rightarrow r_{1,2} = \pm 2i$ → $y_1 = \cos 2t$, → $y_2 = \sin 2t$

$y = c_1 \cos 2t + c_2 \sin 2t$ → general soln

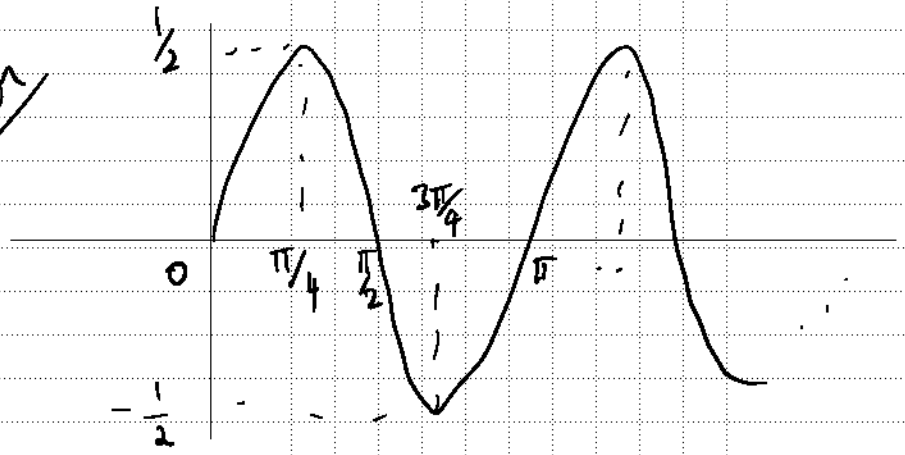
$y' = -2c_1 \sin 2t + 2c_2 \cos 2t$

IVs: $y(0) = 0 \Rightarrow c_1 = 0$

$y'(0) = 1 \Rightarrow 2c_2 = 1 \Rightarrow c_2 = 1/2$

particular soln: $y = \frac{1}{2}\sin 2t$

Sketch



behavior $t \uparrow$

oscillates between $-\frac{1}{2}$ & $\frac{1}{2}$.

In each of Problems 1 through 10, find the general solution of the given differential equation.

3.4.6 $y'' - 6y' + 9y = 0$

char eqn: $r^2 - 6r + 9 = 0$

$(r-3)^2 = 0 \rightarrow r_1, r_2 = 3 \rightarrow y = e^{3t}$, $y_2 = te^{3t}$

general soln: $y = c_1 e^{3t} + c_2 te^{3t}$

How we find $y_2 = te^{3t}$?

We know $c_1 y_1$ is a solution. Instead of c_1 we choose to use a fnc. $v(t)$, and assume $v(t)y_1(t)$ solves the DE. ("Reduction of order")

$y = v(t)e^{3t}$ $\quad y' = v'(t)e^{3t} + 3v(t)e^{3t}$

$y'' = v''(t)e^{3t} + 3v'(t)e^{3t} + 3v'(t)e^{3t} + 9v(t)e^{3t}$

$0 = y'' - 6y' + 9y = e^{3t}[v'' + 6v' + 9v - 6v' - 18v + 9v]$

$0 = v'' \rightarrow v' = c_1 \rightarrow v = c_1 t + c_2$

$y = v \cdot y_1 = (c_1 t + c_2)e^{3t} = c_1 te^{3t} + c_2 e^{3t}$ $\quad y = c_1 y_1 + c_2 y_2$ ($y_1 = e^{3t}$, $y_2 = te^{3t}$)

$$W(y_1, y_2)(t) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{3t} & te^{3t} \\ 3e^{3t} & e^{3t} + 3te^{3t} \end{vmatrix} = e^{6t}(1 + 3t - 3t) = e^{6t}$$

Reduction of order works for 2nd order homogen. linear DE with a known soln.

$y = v \cdot y_1$

$y' = v' \cdot y_1 + v y_1'$

$y'' = v'' y_1 + 2v' y_1' + v \cdot y_1''$

$y'' + p(t)y' + q(t)y = 0$, y_1 soln $\Leftrightarrow y_1'' + py_1' + qy_1 = 0$

$v'' y_1 + 2v' y_1' + v y_1'' + p(v' y_1 + v y_1') + q v \cdot y_1 = 0$

$v'' \cdot y_1 + v'(2y_1' + py_1) + v(y_1'' + py_1' + qy_1) = 0$

$$0 = y'' - 6y' + 9y = y'' - 3y' - 3y' + 9y$$

$$= (y'' - 3y') - 3(y' - 3y)$$

$u'(t)$ $\quad$ $u(t)$

$$0 = u' - 3u$$

$$3u = u' \longrightarrow u(t) = ce^{3t}$$

(c≠0)

$$y' - 3y = ce^{3t} \qquad y' - 3y = 0 \longrightarrow y = e^{3t}$$

ce^{3t} → te^{3t} (use $\mu(x)$ & solve it).

In each of Problems 11 through 14, solve the given initial value problem. Sketch the graph of the solution and describe its behavior for increasing t.

3.4. 12. $y'' - 6y' + 9y = 0, \quad y(0) = 0, \quad y'(0) = 2$

general soln : $y = c_1 e^{3t} + c_2 te^{3t}$

IVs: $y(0) = 0 \rightarrow y(0) = c_1 = 0 \rightarrow y = c_2 te^{3t}$

$y'(0) = 2 \rightarrow y'(t) = c_2 e^{3t} + 3c_2 te^{3t}$

$y'(0) = c_2 = 2$

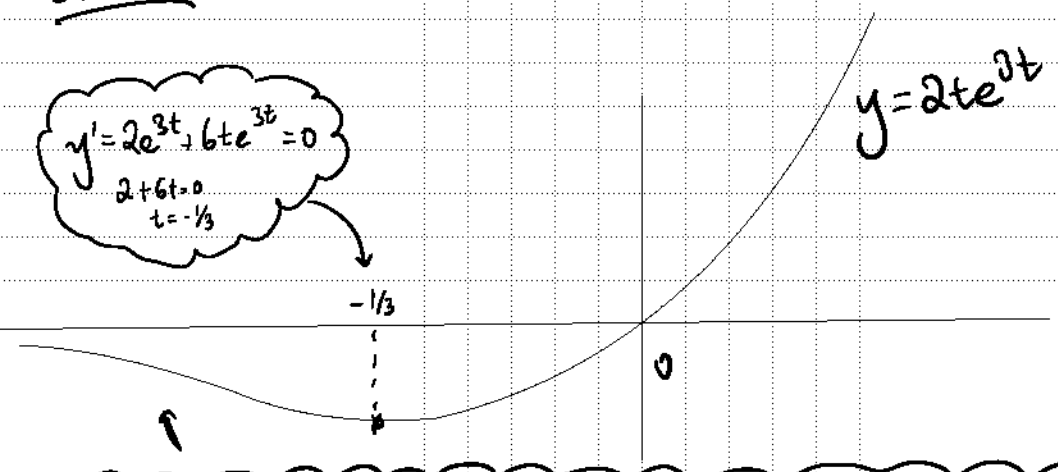
part. soln : $y = 2te^{3t}$

Sketch

$y' = 2e^{3t} + 6te^{3t} = 0$

$2 + 6t = 0$

$t = -1/3$



Behavior

$\lim_{t \to \infty} y = \infty$

y increases exponentially as $t \uparrow$.

since $\lim_{t \to -\infty} y = 0$ (write the limit & use L'Hopital if you don't see it immediately) but $y < 0$ for $t < 0$.

3.4. 18 Consider the initial value problem

$$9y'' + 12y' + 4y = 0, \qquad y(0) = a > 0, \quad y'(0) = -1.$$

(a) Solve the initial value problem.
(b) Find the critical value of a that separates solutions that become negative from those that are always positive.

after some t_0, $\forall t > t_0$ $y(t) < 0$.

(a) 2nd order lin. DE w const. coef.

char eqn: $9r^2 + 12r + 4 = 0$

$(3r+2)^2 = 0 \longrightarrow r_{1,2} = -\frac{2}{3}$

gen. soln: $y = c_1 e^{-\frac{2}{3}t} + c_2 t e^{-\frac{2}{3}t}$ $\qquad y_1 = e^{-2t/3}$ $\qquad y_2 = te^{-2t/3}$

IVs: $y(0) = a \longrightarrow y(0) = c_1 = a > 0.$

$$y = ae^{-\frac{2}{3}t} + c_2 te^{-\frac{2}{3}t}$$

$$y' = -\frac{2}{3}ae^{-\frac{2}{3}t} + c_2 e^{-\frac{2}{3}t} - \frac{2}{3}c_2 te^{-\frac{2}{3}t}$$

$$= e^{-\frac{2}{3}t}\left(-\frac{2}{3}a + c_2\right) - \frac{2}{3}c_2 te^{-\frac{2}{3}t}$$

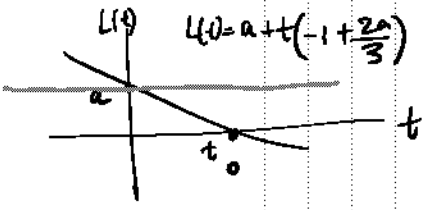
$y'(0) = -1 \longrightarrow y'(0) = -\frac{2}{3}a + c_2 = -1$

$$\Rightarrow c_2 = -1 + \frac{2a}{3}$$

IVs soln: $y = ae^{-\frac{2}{3}t} + \left(-1 + \frac{2a}{3}\right)te^{-\frac{2}{3}t}$

(b) $\lim_{t\to\infty} \underbrace{ae^{-\frac{2}{3}t}}_{\to 0} + \left(-1 + \frac{2a}{3}\right)\underbrace{te^{-\frac{2}{3}t}}_{\to 0} = 0$ nothing interesting.

$$y = \underbrace{e^{-\frac{2}{3}t}}_{>0}\left[\overset{\to 0}{a} + t\left(-1+\frac{2a}{3}\right)\right]$$



as $t\to\infty$ $\quad y<0$ if $-1+\frac{2a}{3}<0 \to a<\frac{3}{2}$

$\quad y>0$ if $-1+\frac{2a}{3}\geq 0 \to a\geq\frac{3}{2}$

critical $a=\frac{3}{2}$.

In each of Problems 23 through 30, use the method of reduction of order to find a second solution of the given differential equation.

3.4. 25. $t^2y'' + 3ty' + y = 0, \quad t > 0, \quad y_1(t) = t^{-1}$

$y = v(t)\,t^{-1}$ $\qquad y = v(t)\,y_1$

$$y' = \frac{v'}{t} - \frac{v}{t^2}$$

$$y'' = \frac{v''}{t} - \frac{v'}{t^2} - \frac{v'}{t^2} + \frac{2v}{t^3}$$

$$t^2\left(\frac{v''}{t} - \frac{v'}{t^2} - \frac{v'}{t^2} + \frac{2v}{t^3}\right) + 3t\left(\frac{v'}{t} - \frac{v}{t^2}\right) + \frac{v}{t} = 0$$

$$tv'' - 2v' + 2\frac{v}{t} + 3v' - 3\frac{v}{t} + \frac{v}{t} = 0 \qquad v'=u,\ v''=u'$$

(1) $tv'' + v' = 0$ $\qquad$ 1st order DE of v'

$tu' + u = 0$

$$\int \frac{v''}{v'}\,dt = \int \frac{-1}{t}\,dt \longrightarrow \ln|v'| = \ln|t|^{-1} + d \qquad |t| = t \text{ since } t>0.$$

$$v' = d\,t^{-1} \qquad e^c \hookrightarrow d$$

(2) $v' = d\,t^{-1}$

$v = d_1 \ln t + c_2 \longrightarrow$ general sol: $y = \frac{v}{t} = c_1 t^{-1}\ln t + c_2 t^{-1}$